

PROBLEM STATEMENT

Assess the Accessibility of Roads, Transport Facilities, and Essential Services

Industry Context

The accessibility of roads, public transport, and essential services plays an important role in improving mobility, safety, and quality of life. People depend on roads and transportation facilities to reach schools, hospitals, workplaces, markets, government offices, and other important locations. However, accessibility can vary significantly between urban and rural areas.

Poor road conditions, inadequate public transportation, traffic congestion, insufficient pedestrian facilities, and limited access to essential services can create difficulties for students, elderly people, persons with disabilities, and economically weaker communities.

These problems may lead to:

- Longer travel times and increased transportation costs
- Difficulty accessing hospitals and emergency services
- Limited access to education and employment
- Increased risk of road accidents
- Difficulties for elderly and differently-abled people
- Unequal access to essential public facilities
- Reduced mobility and quality of life

Existing accessibility assessments are often based on manual surveys or limited datasets. This makes it difficult to obtain a complete and updated understanding of accessibility across different locations. There is a growing need for a systematic approach that evaluates roads, transport facilities, and essential services using location-based and accessibility-related data.

Problem Statement

Design and develop an accessibility assessment system that evaluates the availability and accessibility of roads, transportation facilities, and essential services in a given area.

The system should collect and analyze relevant information such as road connectivity, transportation availability, travel distance, proximity to essential services, and accessibility barriers. It should identify areas where people have limited access to important facilities such as hospitals, schools, public transport stops, markets, and government services.

The solution should provide clear accessibility information through maps, indicators, or visual reports to help identify underserved areas and support better transportation and infrastructure planning.

Objectives

The system should:

- To assess the accessibility and connectivity of roads
- To Evaluate the availability of public transportation facilities
- To Measure the distance to essential services
- To Identify areas with poor accessibility
- To Analyze accessibility barriers for different users
- To Visualize accessibility information using maps or dashboards
- To Identify underserved locations requiring improvement
- To Support data-driven infrastructure and transportation planning

Expected Outcome:

1. Develop an end-to-end Accessibility Assessment System with road accessibility, transport facility, and essential service assessment modules for collecting and analyzing accessibility information.
2. Provide a GUI/dashboard for users to view accessibility information, search locations, and visualize road networks, transport facilities, and essential services.
3. Maintain information such as road connectivity, road conditions, transport facility availability, essential service locations, distance, travel accessibility, and accessibility status for each assessed location.
4. Enable location-based accessibility assessment by evaluating the accessibility of essential services such as hospitals, schools, markets, public transport facilities, and other important services.
5. Identify areas with poor accessibility and underserved locations by analyzing road connectivity, availability of transport facilities, and proximity to essential services.
6. Provide data storage, retrieval, and management functionality for efficiently maintaining road, transport, and essential-service accessibility data.
7. Provide visual reports and accessibility results that allow users to understand accessibility levels, identify gaps, and support infrastructure and transportation planning.
8. Apply key software engineering principles such as security, scalability, maintainability, reliability, and usability throughout the development of the accessibility assessment system.
9. Demonstrate proper requirements analysis, system design, SDLC selection, testing, and evaluation of the system's risks, limitations, performance, and future enhancements.